

7.1.24. If a and y are positive real numbers such that $\log_a y = 2$, what is $\log_{a^2}(y)$? Explain.

(Note, could not use subscript b , so used a)

Definition Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

Let $a, y \in \mathbb{R}^+$

$$\log_a y = 2 \leftrightarrow a^2 = y$$

$$a^2 = y$$

$$\log_{a^2}(y) = x \leftrightarrow (a^2)^x = y$$

$$\log_{a^2}(a^2) = x \leftrightarrow (a^2)^x = a^2$$

$$(a^2)^x = a^{2x}$$

$$a^{2x} = a^2$$

$$x = 1$$

Hence, $\log_{a^2} y = 1$

Solution: $\log_{a^2} y = 1$